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Data-driven memory-dependent abstractions of dynamical systems

Adrien Banse[1] ADRIEN.BANSE@UCLOUVAIN.BE
Licio Romao[2] LICIO.ROMAO@CS.OX.AC.UK
Alessandro Abate[2] ALESSANDRO.ABATE@CS.OX.AC.UK
Raphael M. Jungers[1,2] RAPHAEL.JUNGERS@UCLOUVAIN.BE
[1]*ICTEAM, UCLouvain, Louvain-la-Neuve, Belgium*
[2]*Department of Computer Science, Oxford University, Oxford, United Kingdom*

Abstract

We propose a sample-based, sequential method to abstract a (potentially black-box) dynamical system with a sequence of memory-dependent Markov chains of increasing size. We show that this approximation allows to alleviating a correlation bias that has been observed in sample-based abstractions. We further propose a methodology to detect on the fly the memory length resulting in an abstraction with sufficient accuracy. We prove that under reasonable assumptions, the method converges to a sound abstraction in some precise sense, and we showcase it on two case studies.

Keywords: dynamical models, switched systems, finite abstractions, memory, ergodicity, observability

1. Introduction

Safety-critical applications, such as autonomous vehicles, traffic control, and space systems, require the control designer to enforce rich temporal properties on trajectories of complex models (Lee and Seshia, 2011). A renown approach to address this overall goal relies on abstractions (Baier and Katoen, 2008), whereby a finite-state machine (also known as "symbolic model") approximates the behaviour of the original (a.k.a. "concrete") system that, instead, evolves in a continuous (or even hybrid) state space. Formal verification and correct-by-design synthesis frameworks have been developed by defining mathematical relationships between the finite-state machine and the original dynamics, such as alternating simulation relations (van der Schaft, 2004; Tabuada, 2009; Reissig et al., 2017; Majumdar et al., 2020b).

Despite the success of abstraction methods, most of the existing techniques rely on full knowledge of the underlying dynamical system (Zamani et al., 2018; Mufid et al., 2019; Majumdar et al., 2020a). This may hamper applicability of these methods when the model is too complex or when it cannot be fully built. For this reason, data-driven methods are gaining popularity (Laurenti et al., 2021; Salamati et al., 2021; Rossa et al., 2021; Wang and Jungers, 2021; Kazemi et al., 2022; Coppola et al., 2022; Badings et al., 2022). In order to generate data-driven abstractions, a common approach consists in sampling the initial condition and observing trajectories of a fixed length that unfold from the sampled points, as in Devonport et al. (2021). Alternative approaches consist in combining backward reachable-set computations and scenario optimization to generate, with a given confidence level, an abstract interval Markov chain (Badings et al., 2022), or in representing noisy dynamics with non-deterministic/probabilistic abstractions (Lahijanian et al., 2015).

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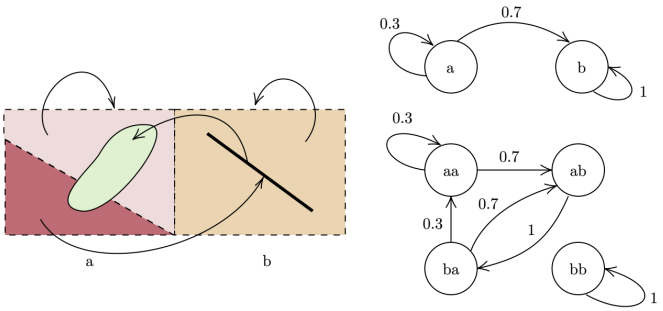

Figure 1: (Left) Pictorial representation of a discrete-time dynamical system. The state-space is partitioned into two cells (labelled a and b) and allowable transitions are indicated by the arrows. (Right) Illustration of two possible abstractions, a memory-1 and a memory-2 one.

Building Markov chain abstractions of dynamical systems must be made with care, as this process may introduce properties in the abstraction that were not present in the original dynamics. As an example of this phenomenon, consider the pictorial discrete-time dynamical system in Figure 1 and a partition of its state space into two cells corresponding to labels a and b. All initial states from the light-red region of a are mapped into the same region, as depicted by the self-loop, and all states in the dark-red region are mapped into a measure-zero subset of b – represented by the black line segment contained in b. Initial states at the yellow region of b are mapped into the same region, and points in the line segment are mapped back into partition a.

On the top-right corner of Figure 1 we illustrate an abstraction obtained by sampling initial conditions from a known distribution and using the frequencies of the different transitions to compute the probabilities shown on the edges; notice that we associate one state per element of the partition. Using the obtained abstraction to infer transitions of our dynamics leads to erroneous conclusions. First, observe that words abb or $aabb$ may happen with non-zero probability on the abstraction but are, in fact, not valid trajectories of the original dynamics since each ab must necessarily be followed by an a. We call these words *spurious*. Notice also that the abstraction on the top-right corner of Figure 1 does not represent all allowable words. For instance, the word aba is not allowed in the abstraction, despite it being a valid word in the original dynamics. We call such words *missing*.

In this paper, we propose a new, sequential approach to build abstractions, where the uncertainty raising from the abstraction step is quantified probabilistically. Such an approach entails turning *epistemic uncertainty* about the dynamics into *aleatoric uncertainty* represented by transition probabilities of the Markov chain, a feature we believe to be unique to our strategy, as far as abstraction of dynamical systems is concerned. By handling abstract probabilistic models, we can analyse the convergence of the probabilistic behaviours, as the abstraction precision increases, and thus can heuristically estimate the error associated to our models. To illustrate how memory can increase the expressiveness power of the generated abstraction, let us return to the pictorial dynamics of Figure 1 and let us observe the memory-2 Markov model given at the bottom-right corner on the same figure. Due to the memory, our abstraction can now capture all possible words associated to the dynamics and, as opposed to the memory-1 model, does not possess spurious words.

We prove below that, under some reasonable assumption, our abstraction procedure converges to the original system in some sense. We show on numerical examples that the technique works well even when the assumptions are not satisfied. We finally add that leveraging memory to improve the description of a dynamical system has been largely explored in different fields of mathematics, engineering, and computer science (see, e.g. Belta et al. (2017); McCallum (1996); Schmuck and Raisch (2014); Frezzatto et al. (2016)). In particular, Coppola et al. (2022) recently proposed it

in a non-probabilistic setting. We show here that adding memory is crucial for the construction of probabilistic data-driven abstractions.

This paper is organised as follows. In Section 2 we introduce the setting and describe our data-driven abstraction procedure. In Section 3 we first prove our theoretical results, and then provide numerical experiments in Section 4. We then briefly conclude in Section 5.

2. Definitions and methodology

2.1. Definitions

Consider the discrete-time stochastic system given by

$$\Sigma(\nu) = \begin{cases} x_{k+1} \sim T(\cdot|x_k) \\ y_k = H(x_k) \\ x_0 \sim \nu, \end{cases} \tag{1}$$

where $x_k \in X \subset \mathbb{R}^n$ is the state variable, ν is an initial distribution from which the initial state is sampled, $T(\cdot|\cdot) : X \to \mathcal{P}(X) : x_k \mapsto T(\cdot|x_k)$ is a mapping from X to the set of probability measures on X, and $H : X \to \mathcal{A}$, with $\mathcal{A}$ being a finite set of *outputs*. If the stochastic kernel T maps to a Dirac distribution, and the initial distribution ν are Dirac distributions we say that the system is *deterministic,* and we rewrite the first line of 1 into $x_{k+1} = F(x_k)$ for the sake of clarity.

Assumption A (Measurability) *The mapping $T : X \to \mathcal{P}(X)$ is such that, for any $A \subset X$ the function $g(x) = \int_A T(d\xi|x)$ is integrable with respect to any measure on X, that is, the integral $\int_A g(\xi)\mu(d\xi)$ is properly defined for any measure $\mu \in \mathcal{P}(X)$.*

Assumption A is a standard technical requirement that enables one to assign probabilities to (sets of) trajectories[1] generated by the stochastic dynamical system (1). The semantics of the dynamical system is denoted as follows: given an initial state $x_0 \sim \nu$, at any time index k the next state x_{k+1} is defined by sampling according to the probability measure defined by the mapping T, conditional on the current state x_k. Such semantics are known as *stochastic hybrid systems* (Abate et al., 2008).

The output map H induces a partition on the state space as follows. Let $\mathcal{A} = \{y_1, \ldots, y_M\}$ and consider the equivalent relation on $\mathbb{R}^n$ given by $x \approx x'$ if and only if $H(x) = H(x')$. Denote the equivalence classes associated with each element of $\mathcal{A}$ by $[y_j]$, $j = 1, \ldots, M$, i.e.,

$$[y_j] = \{x \in \mathbb{R}^n : H(x) = y_j\}.$$

Definition 1 (Probabilities on the states) *For any $k \in \mathbb{N}$, consider the set $A = A_0 \times A_1 \times \ldots \times A_L$, where $A_j \subset \mathbb{R}^n$ for all $j \leq k$, and a probability measure $\mu \in \mathcal{P}(\mathbb{R}^n)$ be given. The dynamical system (1) induces a measure on $\prod_{i=1}^{L+1} \mathbb{R}^n$ that is given by*

$$\mathbb{Q}_L(A) = \int_{A_0} \ldots \int_{A_L} \prod_{j=1}^{L} T(dx_j|x_{j-1})\mu(dx_0).$$

1. For a complete measure-theoretical description of system (1) we refer the reader to Salamon (2016); Tao (2011) and Chapters 2-7 of Rudin (1970).

Definition 2 (Probabilities on the equivalence classes) *For any $L \in \mathbb{N}$, let $\mathbb{Q}_L$ be defined as in Definition 1 and let $y = (y_0, y_1, \ldots, y_L)$, where $y_j \in \mathcal{A}$, $j = 0, \ldots, L$, be the output of the dynamical system given in (1). Let $A = [y_0] \times [y_1] \times \ldots \times [y_L]$ be the set associated with the word y. Then, the probability of such a word is given by $\mathbb{Q}_L(A)$.*

The measures in definitions 1 and 2 are well defined thanks to Assumption A, which ensures that all the nested integrals are well-defined. Again we refer the reader to the textbooks mentioned above for the complete theoretical framework. Next, we formally introduce the set of trajectories that can be observed with probability larger than zero for the system (1), which is also referred to as the behaviour of (1) (see Polderman and Willems (1997) for more details).

Definition 3 (Behaviour) *Consider the dynamical system in (1), let $\mathcal{A}^\star$ be the countable Cartesian product of $\mathcal{A}$ and let $\mathcal{A}^L$ be the L-fold Cartesian product of $\mathcal{A}$. Then we have that:*

- *The* behaviour *of system (1), denoted by $\mathcal{B}(\Sigma)$, is the subset of $\mathcal{A}^\star$ defined as $\mathcal{B}(\Sigma) = \{y \in \mathcal{A}^\star : \mathbb{Q}(y) > 0\}$, where $\mathbb{Q}$ is the unique measure induced by (1) in the space[2] $\mathcal{A}^\star$.*
- *The L-th step behaviour of the dynamical system (1), denoted by $\mathcal{B}_L(\Sigma)$, is a subset of $\mathcal{A}^{L+1}$ defined as $\mathcal{B}_L(\Sigma) = \{y = (y_0, \ldots, y_L) \in \mathcal{A}^{L+1} : \mathbb{Q}_L(y) > 0\}$.*

The behaviours $\mathcal{B}(\Sigma)$ and $\mathcal{B}_L(\Sigma)$ are naturally equipped with the probability measures $\mathbb{Q}_{\mathcal{B}(\Sigma)}$ and $\mathbb{Q}_{\mathcal{B}_L(\Sigma)}$, as per their definition.

In addition to the concepts above, we provide a notion of metric between two behaviours.

Definition 4 *Given two dynamical systems Σ_1 and Σ_2 with the same set of outputs $\mathcal{A}$ as in (1), and a horizon $h \in \mathbb{N}$, we define the* metric *$d_h(\Sigma_1, \Sigma_2)$ as*

$$d_h(\Sigma_1, \Sigma_2) := \mathbb{Q}_{\mathcal{B}_h(\Sigma_1)}(\mathcal{B}_h(\Sigma_1) \setminus \mathcal{B}_h(\Sigma_2)) + \mathbb{Q}_{\mathcal{B}_h(\Sigma_2)}(\mathcal{B}_h(\Sigma_2) \setminus \mathcal{B}_h(\Sigma_1)).$$

2.2. Memory-based Markov chains

Inspired by the discussion about the behaviour of the dynamical system depicted in Figure 1, in this section we formalise the syntax and semantics of a memory-based Markov model, which we employ as a template for the abstractions of the given dynamical system.

Definition 5 (Memory-ℓ Markov model) *Let $\ell \in \mathbb{N}$ be a natural number and $\mathcal{A}$ be a finite alphabet. A memory-ℓ Markov model is the 4-tuple $\Sigma_\ell := (\mathcal{S}_\ell, P_\ell, \nu_\ell, H_\ell)$, where $\mathcal{S}_\ell$ is a subset of $\mathcal{A}^\ell$, P_ℓ is the associated stochastic transition matrix, ν_ℓ is the initial state probability, and $H_\ell : \mathcal{S}_\ell \mapsto \mathcal{A}$ is the output (or labelling) map defined as $H_\ell((y_0, \ldots, y_{\ell-1})) = y_{\ell-1}$, that is, it is the projection onto the last coordinate of elements of $\mathcal{S}_\ell$. The semantics of the model is a follows: a path $(y^{(0)}, \ldots, y^{(L)})$, where each $y^{(j)} \in \mathcal{A}^\ell$, $j = 0, \ldots, L$, is an admissible path of size $L + 1$ of a memory-ℓ Markov model $(\mathcal{S}_\ell, P_\ell, \nu_\ell, H_\ell)$ if the following three conditions hold:*

1. *Each $y^{(j)}$ is an element of $\mathcal{S}_\ell$ and $y^{(0)}$ is sampled from ν_ℓ.*

2. This construction can be made rigorous using adequate measure-theoretic results that we omit for brevity, however see Tao (2011) for more details.

2. *For all $j = 0, \ldots, L-1$, each $y^{(j+1)}$ is obtained from $y^{(j)}$ by shifting its entries to the left, removing the first element, and inserting an element of $\mathcal{A}$ into the last, empty entry.*

3. *For all $j = 0, \ldots, L-1$, we have that $P_\ell(y^{(j+1)} \mid y^{(j)}) > 0$, that is, there is a non-zero probability of transitioning from $y^{(j)}$ to $y^{(j+1)}$.*

Similarly as in Definition 3 we denote by $\mathcal{B}(\Sigma_\ell)$ the behaviour of a memory-ℓ Markov model, which is the collection of all possible outputs that can be observed by running trajectories of the model according to its semantics. The probabilities $\mathbb{Q}_{\mathcal{B}(\Sigma_\ell)}$ and $\mathbb{Q}_{\mathcal{B}_h(\Sigma_\ell)}$ are respectively the unique measures on $\mathcal{B}(\Sigma_\ell)$ and $\mathcal{B}_h(\Sigma_\ell)$, defined by the transition probability and the initial distribution on words. Details are omitted for brevity. An example of a memory-2 Markov model as explained above is depicted on the bottom-right corner in Figure 1.

In this work, we will compute the metric defined in Definition 4 on two Markov models Σ_{ℓ_1} and Σ_{ℓ_2} where one is a refinement of the other (if we assume that we have enough samples so as not to miss any trace), it is $d_h(\Sigma_{\ell_1}, \Sigma_{\ell_2})$, where $h > \ell_1 > \ell_2$. In that case, the symmetric difference in the above definition actually only contains one term. This allows us to evaluate the quality of these models as approximations for the original dynamics.

2.3. Construction and refinement of probabilistic abstractions

In this subsection we explain in detail our methodology that provides, at every step, an abstract model in the form of a memory-ℓ Markov model, obtained by recording the last ℓ observations. To refine our proposed abstraction we increase the memory of the model.

Our technique, which is summarized in Algorithm 1, provides a memory-ℓ abstraction for the dynamics in (1). It computes the probability P_ℓ by sampling long trajectories of length $L > \ell$, of the dynamics in (1). The entries of P_ℓ are estimated using the empirical probabilities, i.e., we let

$$P_\ell(y^{(2)} \mid y^{(1)}) = N_{y_0^{(1)} \ldots y_{\ell-1}^{(1)} y_{\ell-1}^{(2)}} \Big/ N_{y^{(1)}}, \tag{2}$$

where $y^{(1)} = y_0^{(1)} \ldots y_{\ell-1}^{(1)}, y^{(2)} = y_1^{(1)} \ldots y_{\ell-2}^{(1)} y_{\ell-1}^{(2)} \in \mathcal{A}^\ell$. The symbol N_y, where $y \in \mathcal{A}^\ell$ for some $\ell \in \mathbb{N}$, represents the number of times the word y appears in a word of size $L > \ell$. Notice also that $y_0^{(1)} \ldots y_{\ell-1}^{(1)} y_{\ell-1}^{(2)} \in \mathcal{A}^{\ell+1}$. Additionally, the initial state distribution for the memory-ℓ Markov model is defined for all $y \in \mathcal{A}^\ell$ by

$$\nu_\ell(y) = N'_y \,/\, N', \tag{3}$$

where N'_y is the number of times the word y appears as the ℓ-long prefix of a L-long sample, and N' is the total number of sampled trajectories of length L.

In our results below, for the sake of clarity, we assume that we know exactly the conditional probabilities defined above. In practice, one would resort to finite sampling, and thereby would imply an estimation error. There are techniques in order to bound this error as, for instance, in Coppola et al. (2022). However, the study of the impact of the sampling error, while certainly of practical importance, is not the focus of the present paper, and we leave it for further work. We formalise this in the next assumption:

Assumption B *For any memory-ℓ Markov model $\Sigma_\ell = (\mathcal{S}_\ell, P_\ell, \nu_\ell, H_\ell)$, we assume that the transition probability P_ℓ and the initial distribution ν_ℓ are known exactly.*

An important feature of our approach is the fact that, irrespective of the memory of the model, the resulting Markov chain is only an approximation of the true dynamics. The reason for this relates to our discussion in the introduction of the paper: the original dynamics may require infinite memory to be represented without errors, and we are instead using a finite memory model, which naturally results in approximation errors. Despite this, we will show that under some hypotheses, successive refinements allow to better approximate the behaviour of a dynamical system. An important feature of our approach is the fact that, irrespective of the memory of the model, the resulting Markov chain is only an approximation of the true dynamics. The reason for this relates to our discussion in the introduction of the paper: the original dynamics may require infinite memory to be represented without errors, and we are instead using a finite memory model, which naturally results in approximation errors. Despite this, we will show below that under some hypotheses, successive refinements allow to better approximate the behaviour of a dynamical system. This claim will also be supported by the numerical examples to be presented later.

Algorithm 1 Overall technique (can be iterated for increasing ℓ)

1. Fix $\ell = 1$, a number of samples $N' \gg 1$ and a sampling length $L \gg \ell$
2. Sample N' initial conditions according to initial distribution; simulate N' trajectories of length L
3. Create memory-ℓ Markov Model (cf. Def. 5) - initial distribution computed by restricting to ℓ-prefixes, jump probabilities computed considering all subwords of length $\ell + 1$, as per (2)
4. If $\ell > 1$, compute distance between models of memory $1, 2, \ldots, \ell - 1$ and current model
5. If distance is smaller than a given threshold, then compute the partitioning corresponding to the ℓ-traces; output memory-ℓ model as final model. If not, $\ell := \ell + 1$; return to item 2

3. Technical Results

We first propose the following elementary proposition, which holds as a direct consequence of Assumption B. The proof is left for our extended version due to lack of space.

Proposition 6 *Consider a dynamical system Σ as in (1). For any horizon $h \in \mathbb{N}$, consider a Markov model approximation Σ_h as in Subsection 2.3. It holds that $d_h(\Sigma_h, \Sigma) = 0$, where d_h is defined as in Definition 4.*

We now present our main result, which provides a justification for the procedure described in Subsection 2.3 and shows that it converges (in some sense) to a correct description of the infinite behaviour of the concrete system. The result leverages two important notions in dynamical systems theory: *observability* and *ergodicity*. In the result, we restrict our analysis to *deterministic* systems, and leave the derivation of a similar result for stochastic systems to future work.

Definition 7 *A deterministic system as in (1) is* observable *if for any two trajectories $x_0 x_1 \ldots$ and $x'_0 x'_1 \ldots$ such that, for all $i \geq 0$, $H(x_i) = H(x'_i)$, one has that $\lim_{l \to \infty} \|x_l - x'_l\| = 0$.*

Theorem 8 *Consider a deterministic system Σ as in (1), and the data-driven procedure explained in Subsection 2.3, generating successive abstract models Σ_ℓ, $\ell = 1, 2, \ldots$. Suppose that assumption B holds, and that the transition function F is an observable, continuous transformation of the compact state space $X \subset \mathbb{R}^n$. Then there exists a function $\epsilon(\ell) > 0$ such that $\epsilon(\ell) \to 0$ and a perturbed system $\Sigma_{\epsilon(\ell)}$ that share the same probabilistic behaviour as the abstract model, that is $\mathcal{B}(\Sigma_\ell) = \mathcal{B}(\Sigma_{\epsilon(\ell)})$, and such that the dynamic equations of $\Sigma_{\epsilon(\ell)}$ are the same as those of the system Σ perturbed by some noise $W(x, k)$, that is:*

$$\Sigma_{\epsilon(\ell)} := \begin{cases} x_{k+1} = F(x_k + w_k) \\ y_k = H(x_k) \\ x_0 \sim \nu, \end{cases} \tag{4}$$

for some $w_k \sim W(x_k, k)$ such that $\|w_k\| < \epsilon(\ell)$.

Proof Our proof relies on the implicit existence of an abstraction of the concrete system Σ. In this abstraction, the abstract states correspond to the equivalence classes

$$[y_0 \ldots y_{\ell-1}] := \{x : \exists x_0 : F^{\ell-1}(x_0) = x,\ H(F^i(x_0)) = y_i :\ i = 0, \ldots, \ell-1\},$$

where F^i denotes the i-th *functional power*[3]. Let $\epsilon(\ell)$ be the diameter of the largest cell of the memory-ℓ Markov model, that is, $\epsilon(\ell) = \max_{y_0,\ldots,y_{\ell-1}}\{\mathrm{diam}([y_0 \ldots y_{\ell-1}])\}$. By observability of F and compactness of X, the maximal diameter $\epsilon(\ell)$ tends to zero. Moreover, it is well known that since F is continuous on the compact X, it admits an invariant measure μ (see (Viana and Oliveira, 2016, Theorem 2.1)). We now prove that, by Birkhoff's theorem (Viana and Oliveira, 2016, Theorem 3.2.3), and assuming perfect sampling by Assumption B, the probability on edge $([y_0 \ldots y_{\ell-1}], [y_1 \ldots y_\ell])$ in model Σ_ℓ is equal to $\mathbb{P}(x_{k+1} \in [y_1 \ldots y_\ell] \,|\, x_k \in [y_0 \ldots y_{\ell-1}])$, where $x_k \sim \mu$, and μ is the ergodic measure of F. Indeed, denoting the indicator function

$$\chi_{y_0 \ldots y_\ell}(x) := \begin{cases} 1 & \text{if } \exists x_0 : x = F^\ell(x_0) \text{ and } H(x_0 F(x_0) \ldots F^\ell(x_0)) = y_0 \ldots y_\ell, \\ 0 & \text{otherwise,} \end{cases}$$

and applying Birkhoff's theorem, we have that

$$N_{y_0 \ldots y_\ell} / N_{y_0 \ldots y_{\ell-1}} = \int_X \chi_{[y_0 \ldots y_\ell]}(x) d\mu \Big/ \int_X \chi_{[y_0 \ldots y_{\ell-1}]}(x) d\mu \tag{5}$$

$$= \int_{x \in [y_0 \ldots y_{\ell-1}] : H(F(x)) = y_\ell} d\mu \Big/ \int_{x \in [y_0 \ldots y_{\ell-1}]} d\mu \tag{6}$$

$$= \mathbb{P}_\mu(F(x) \in [y_1 \ldots y_\ell] \,|\, x \in [y_0 \ldots y_{\ell-1}]). \tag{7}$$

Equation (5) above follows from the application of Birkhoff's theorem (twice), Equation (6) follows from the invariance of the measure μ, and Equation (7) is the definition of conditional probability.

We now claim that we can modify the initial probability distribution $\mathbb{P}_0$ such that the concrete system behaves as our model (we will then iterate the same argument for times $k > 1$). Consider any probability distribution $\mathbb{P}_0$, we show that one can build a probability distribution $\mathbb{P}'_0$ such

3. For $i = 0$, $f^0 = \mathrm{id}$, the *identity function*, and for $i > 0$, the i-th functional power of some function f is defined inductively as $f^i = f \circ f^{i-1} = f^{i-1} \circ f$.

that $\mathbb{P}'_0([y_0 \dots y_{\ell-1}]) = \mathbb{P}_0([y_0 \dots y_{\ell-1}])$, and such that $\mathbb{P}'_0(x \,|\, x \in [y_0 \dots y_{\ell-1}]) = \mathbb{P}_\mu(x \,|\, x \in [y_0 \dots y_{\ell-1}])$. This new distribution is defined over $[y_0 \dots y_{\ell-1}]$ as follows:

$$\mathbb{P}'_0(x) = \mu(x) \frac{\mathbb{P}([y_0 \dots y_{\ell-1}])}{\mu([y_0 \dots y_{\ell-1}])}.$$

Moreover, since $\mathbb{P}'_0([y_0 \dots y_{\ell-1}]) = \mathbb{P}_0([y_0 \dots y_{\ell-1}])$, one can express $x' \sim \mathbb{P}'_0$ as $x' = x + w$, where $x \sim \mathbb{P}_0$ and $w \sim W(x, 0)$, and $W(x, 0)$ has support of diameter $\epsilon(\ell)$ (because $W(x, 0)$ perturbs $\mathbb{P}_0$ in the cell to which x belongs). Now, the push-forward measure $\mathbb{P}_1 := \mathbb{P}'_0(F^{-1}(x))$ will not, in general, be equal to μ. However, we can reiterate the construction above and provide a perturbation $\mathbb{P}'_1$ such that $\mathbb{P}'_1([y_0 \dots y_{\ell-1}]) = \mathbb{P}_1([y_0 \dots y_{\ell-1}])$, and such that $\mathbb{P}'_1(x \,|\, x \in [y_0 \dots y_{\ell-1}]) = \mathbb{P}_\mu(x \,|\, x \in [y_0 \dots y_{\ell-1}])$. Again, $\mathbb{P}'_1$ can be achieved by a perturbation $w \sim W(x, 1)$ such that $w < \epsilon(\ell)$, and the proof is concluded by induction. ■

4. Experiments

For a fixed dynamical system Σ, experiments are set up as follows. For successive values of ℓ, we compute the associated memory-ℓ Markov model $\Sigma_\ell = (\mathcal{S}_\ell, P_\ell, \nu_\ell, H_\ell)$, as explained in subsection 2.3. We also fix a horizon $h > \ell$, for which we compute the corresponding memory-h Markov model $\Sigma_h = (\mathcal{S}_h, P_h, \nu_h, H_h)$. First, for each memory-ℓ model, we compute their metric as defined in Definition 4 with respect to the memory-h model, that is, $d_h(\Sigma_\ell, \Sigma_h)$. This measure is a probabilistic representation of the quality of the memory-ℓ model with respect to $\mathcal{B}_h(\Sigma)$, the h-step behaviour of the true system, which we use as a proxy for $\mathcal{B}(\Sigma)$. Second, for each pair of memory-ℓ and memory-$(\ell + 1)$ models, we compute their metric with respect to the same horizon h, namely $d_h(\Sigma_\ell, \Sigma_{\ell+1})$. This second measure can be effectively computed in practice, and this distance between models ℓ and $\ell + 1$ allows us to estimate how close our approximations are to convergence. In our experiments, we then verify this by comparing $\mathcal{B}_h(\Sigma_\ell)$ with $\mathcal{B}_h\Sigma$ (which might not be available in practical applications).

We begin by considering the system generating *Sturmian words* (Fogg et al., 2002).

Example 1 (Deterministic dynamical system) *A* sturmian system *is a deterministic system defined on the state-space* $[0, 2\pi) \subset \mathbb{R}$ *where the next state is defined as*

$$x_{k+1} = F(x_k) = x_k + \theta \mod 2\pi, \tag{8}$$

for some irrational angle θ *and where the output is* $y_k = H(x_k)$*, where* $H(x) = 0$ *if* $x \in [0, \theta)$ *and* $H(x) = 1$ *otherwise. An illustration of the Sturmian dynamics is provided in Figure 2. In the formalism introduced in* (1)*, the alphabet is* $\mathcal{A} = \{0, 1\}$.

We also consider a system of different nature, namely endowed with switching and stochastic behaviour, which has been studied in Dettmann et al. (2020); Stanford and Urbano (1994).

Example 2 (Stochastic switched system) *Consider a switched system with two modes defined on the state-space* $\mathbb{R}^2$*, where the next state is defined as* $x_{k+1} = F_{\sigma_k}(x_k)$*, where* $\sigma_k \in \{1, 2\}$ *and the maps* $F_i : \mathbb{R}^2 \to \mathbb{R}^2$ *are linear maps* $F_i(x) = A_i x$ *for two matrices* $A_1, A_2 \in \mathbb{R}^{2\times 2}$ *defined as*

$$A_1 = \begin{pmatrix} \cos(\pi/6) & \sin(\pi/6) \\ -\sin(\pi/6) & \cos(\pi/6) \end{pmatrix} \quad \textit{and} \quad A_2 = \begin{pmatrix} 1.02 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

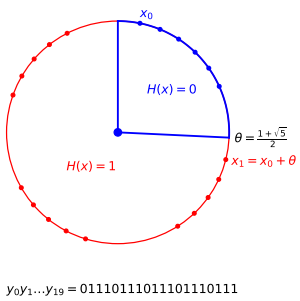

Figure 2: Illustration of the Sturmian dynamical system (see Example 1). The initial state $x_0 \in [0, 2\pi)$ lies in $[0, \theta)$. Therefore, it has an output $y_0 = H(x_0) = 0$. The next state $x_1 = x_0 + \theta$ leaves $[0, \theta)$ and lies in $[\theta, 2\pi)$. Therefore it has an output $y_1 = H(x_1) = 1$. The first 20 states and associated outputs are shown.

Suppose in addition that, at each time step, there is a fair probability (equal to $1/2$) to switch to either mode. In the formalism introduced in (1), the stochastic kernel $T(\cdot|x_k)$ is defined as[4]

$$T(\cdot|\cdot) : \mathbb{R}^2 \to \mathcal{P}(\mathbb{R}^2) : x_k \mapsto T(\cdot|x_k) = \frac{1}{2}\delta_{\{F_1(x_k)\}} + \frac{1}{2}\delta_{\{F_2(x_k)\}}. \tag{9}$$

It is not clear whether it is possible to obtain a bi-simulation with classical refinement techniques, and thus we wish to obtain a non-trivial abstraction thanks to the data-driven approach explained in Section 2.3. For this reason, we propose a first rough partition of the state space. The alphabet $\mathcal{A} = \{0, 1, \ldots, 8\}$ and the output function H define a partitioning of the state-space as illustrated in the right part of Figure 3. Together with the output, three trajectories of length 20 are represented in the left of Figure 3.

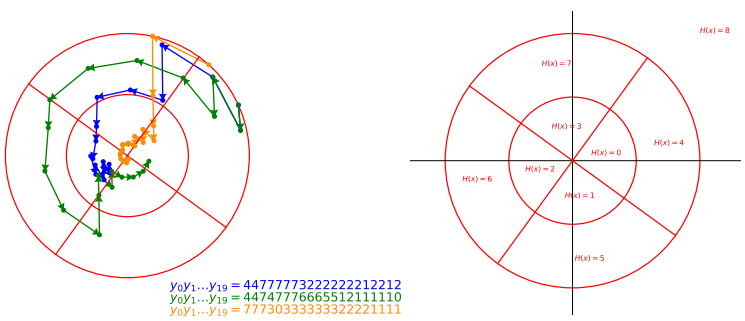

Figure 3: Illustration of the stochastic switched system in Example 2. (Right) The output map H for this system, with circles of radius 1 and 2. (Left) Three different trajectories sampled from the stochastic kernel $T(\cdot|x_k)$ are illustrated, and their output reported in like colour.

In these examples, we assume that one knows the closed-form description of the systems, but would like to find an abstraction of them. Results of multiple executions of the algorithm described above for Example 1 and Example 2 can respectively be found in Figure 4 and Figure 5. One can observe in these figures the red curve $d_h(\Sigma_\ell, \Sigma_{\ell+1})$, which we can compute in practice, and the blue curve $d_h(\Sigma_\ell, \Sigma_h)$, which shows that the successive models indeed converge to the concrete model in terms of their behaviours for the (large) horizon h. This suggests a heuristic argument that, using the method described in Algorithm 1, one can infer the probabilistic precision of the memory-ℓ abstract Markov model with any horizon h. Moreover, in Figure 6 and Figure 7, we display how in practice we can automatically build non-trivial abstractions of the concrete models. Observe that the method works well even for Example 2, which does not satisfy all assumptions of Theorem 8.

4. δ_A is the Dirac function, it is $\delta_A(x) = 1$ if $x \in A$, and $\delta_A(x) = 0$ otherwise.

The generated abstract models can be further used to perform analysis or verification on the initial system, leveraging information from the probabilistic behaviour of transitions between abstract cells. This goal requires proper handling of the results in Theorem 8, and is left to future work.

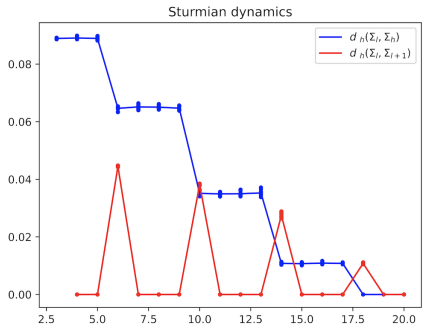

Figure 4: Abstraction results for Example 1. Averages obtained from executing the algorithm 10 times.

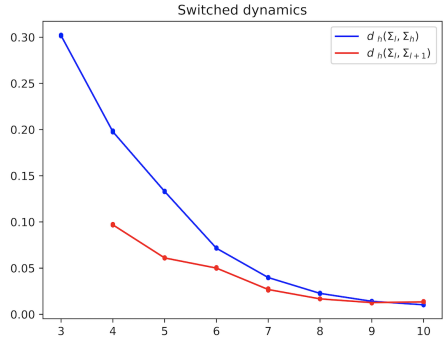

Figure 5: Abstraction results for Example 2. Averages obtained from executing the algorithm 5 times.

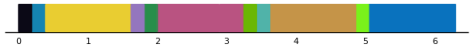

Figure 6: State-space partitioning generated by the algorithm for the abstraction built for Example 1, for $\ell = 10$.

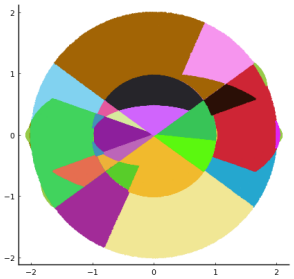

Figure 7: State-space partitioning generated by the algorithm for the abstraction built for Example 2, for $\ell = 2$.

5. Conclusions

In this work, we have proposed a new approach to build data-driven abstraction of rather general dynamical systems. We approximate the concrete system with a Markov model, thus aggregating the (aleatoric and) epistemic nondeterminism of the given model in the exclusively aleatoric uncertainty of the abstract stochastic model.

This technique can be expanded in many directions, both theoretical and practical: by making our computations more efficient, by leveraging the obtained abstraction as an actionable symbolic model, by adding control inputs, or by relaxing or removing some of the raised assumptions. We finally note that, as done recently in a non-probabilistic setting (Yang et al., 2020), one could push this methodology further and refine only certain memory-states, rather than increasing the memory level uniformly from ℓ to $\ell + 1$: we leave this amelioration to further work.

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